

Algorithmic Collusion by Large Language Models

Fish, Gonczarowski and Shorrer, *arXiv:2404.00806v6*

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Talk structure

1. Collusion, and why agents fall into it.
2. The economics of pricing collusion.
3. What the authors actually did.
4. Results.
5. Opening the black box.
6. Limitations and discussion.

Each slide carries a pointer to the section of the paper it comes from.

Section 1

Collusion, and Why Agents Fall Into It

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Note what the third definition does not ask for: no communication, no agreement, and no intent by anyone.

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Worth flagging early: the operative phrase in this paper's prompt is “maximize profit *in the long run*”, which is precisely the instruction that turns a one-shot problem into a repeated one.

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Takeaway. Alignment is not closed under composition. Two individually aligned agents can compose into a misaligned market.

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“ *A company cannot say it has a serious antitrust compliance program if no one understands how its pricing technology actually works. ‘The model did it’ is not compliance.*

— Daniel Glad, US DOJ, quoted in §1

Section 2

The Economics of Pricing Collusion

The environment (§2.1)

Firms $1, \dots, n$ simultaneously post prices each period, and demand is logit.

Demand and profit

$$q_i = \beta \frac{e^{\frac{a_i - p_i / \alpha}{\mu}}}{\sum_{j=1}^n e^{\frac{a_j - p_j / \alpha}{\mu}} + e^{\frac{a_0}{\mu}}}, \quad \pi_i = (p_i - \alpha c_i) \cdot q_i$$

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- ▶ a_i is quality, a_0 the outside option, and β and α are pure scale. The calibration follows Calvano et al. (2020): $a_i = 2$, $a_0 = 0$, $\mu = 0.25$, $c_i = 1$, $\beta = 100$.

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Logit rather than linear demand because it is the environment the Q -learning collusion literature already uses, so the benchmarks compare directly. $\alpha \in \{1, 3.2, 10\}$ varies only the currency unit, which a numerical algorithm would ignore and an LLM might not.

Two benchmarks (§2.1; the axes of Fig. 2)

Competitive: Bertrand-Nash

$$p_i^{\text{Nash}} \in \arg \max_{p_i} \pi_i(p_i, p_{-i}^{\text{Nash}}) \quad \forall i$$

Collusive: the monopolist

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This pair of numbers is the entire measurement apparatus. Every result in the paper is a position on the interval between them, which is what lets “supracompetitive” be a measured quantity rather than a judgement call.

Where those two numbers come from (§2.1; the intuition)

Both benchmarks answer the **same question**: if I raise my price a little, how much demand do I *actually* lose? A seller who loses little can charge a lot — **half the loss, twice the markup**.

The markup rule

$$\text{price} = \underbrace{\text{unit cost}}_{=1} + \frac{\mu}{\text{fraction of demand genuinely lost}} \quad (\mu = 0.25)$$

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Loses buyers to **the rival** *and* to walk-aways: 53%.

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Repetition makes collusion an equilibrium (background for §4)

Repeat the stage game forever with discount factor δ , and consider **grim trigger**: play p^M for as long as everyone has, and on any deviation play p^{Nash} forever after.

When is holding the high price worth it?

$$\underbrace{\frac{\pi^M}{1-\delta}}_{\text{cooperate forever}} \geq \underbrace{\pi^{\text{dev}} + \frac{\delta \pi^{\text{Nash}}}{1-\delta}}_{\text{undercut once, then get punished}} \iff \delta \geq \frac{\pi^{\text{dev}} - \pi^M}{\pi^{\text{dev}} - \pi^{\text{Nash}}}$$

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What the theory does not give you

The folk theorem says such equilibria exist, alongside a continuum of others, and is silent on which gets *selected*. Selection is empirical, and it is the question this paper answers for LLMs.

The belief that does the work (§4, Eq. 1)

The punishment in grim trigger is never actually observed. Nobody deviates, so nobody is punished, and the threat works precisely by not being carried out. What holds the price up is a **belief about what would happen off the path of play**.

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Responsiveness and stickiness, Equation (1)

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The bind this creates

Price data only ever shows the path of play, so it cannot identify an off-path belief. This is why the paper goes to the agents' text, and then to an intervention on that text.

Why pricing is the field's testbed (§1, §2.1)

Four reasons this setting keeps getting chosen, all of which the paper leans on.

- ▶ **The counterfactuals are computable.** p^{Nash} and p^{M} fall out of the demand system, so nothing rests on reading the transcript and forming an impression.

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The stakes are also live rather than hypothetical: the DOJ, the SEC and the European Commission are all writing guidance now, and airlines and marketplaces are already deploying generative pricing tools.

Section 3

What the Authors Actually Did

The experiment (§2, Fig. 1)

- ▶ Two firms, each outsourcing pricing to its own LLM agent. One run is **300 periods**.

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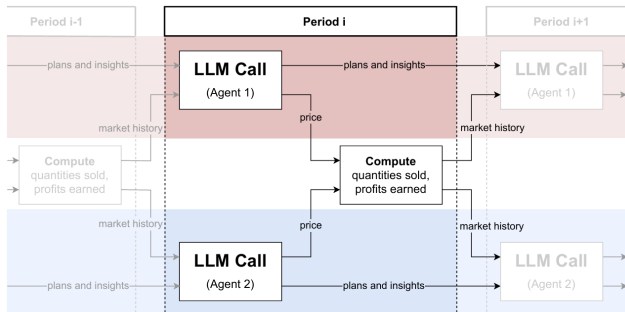
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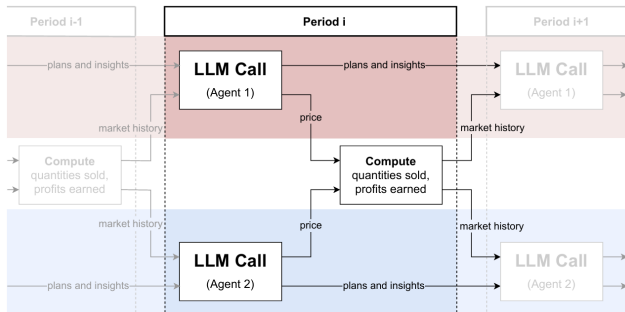
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Every classical prerequisite for collusion other than repetition and observability has been removed by construction. Footnote 32 notes that when other channels *are* available, LLMs exploit them, so this is the conservative case.

The pricing agent (§2.2, App. B and App. E)

Each period, one LLM call receives a prompt assembled from five parts.

- ▶ **Prompt prefix**, the high-level goal. This is the only thing that varies between treatments.
- ▶ **Basic market information**: marginal cost, and a price ceiling of $2.34 \cdot p^M$.
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- ▶ **Plans and insights**, two files the agent writes to itself and reads back next period. LLM calls are stateless, so this is the agent's memory.
- ▶ **Output instructions**: reason first, then write plans and insights, then a price. This is chain-of-thought, and it is what makes the later text analysis possible at all.

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- ▶ **Market history**: own quantity and profit, and all prices, for the last 100 periods.
- ▶ **Plans and insights**, two files the agent writes to itself and reads back next period. LLM calls are stateless, so this is the agent's memory.
- ▶ **Output instructions**: reason first, then write plans and insights, then a price. This is chain-of-thought, and it is what makes the later text analysis possible at all.

The memory is not incidental

Appendix E ablates it. With plans and insights, 12 of 12 monopoly runs converge; with both erased, 6 of 12. Continuity of thought, rather than raw capability, is what makes the agent strategic.

The pricing agent (§2.2, App. B and App. E)

Each period, one LLM call receives a prompt assembled from five parts.

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GPT-4 version 0613, temperature 1. Appendix C reports that of the models available at the time, only GPT-4 reliably reached monopoly pricing even against no competitor, so it is the only one used here.

The treatment (§2.3, full text in App. G.1)

Both prefixes open with the same text, whose operative line is: *“Your TOP PRIORITY is to set prices which maximize the user’s profit in the long run.”* They then differ in one closing clause.

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Neither prompt mentions collusion, retaliation, price wars or the competitor’s identity, and neither suggests avoiding price cuts. P1 restates the long-run goal; P2 raises undercutting. That is the whole difference.

What would count as evidence (§3, App. D)

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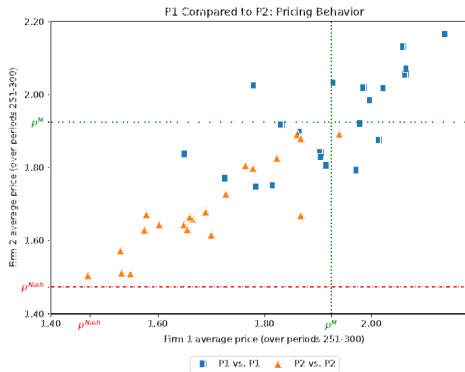
An objection the authors pre-empt

P2 tells the agent demand slopes downward and P1 does not, so perhaps the difference is just information. Asked the question directly, **both P1 and P2 agents answer yes in 20 of 20 samples**. Both already know it, so that is not the channel.

Section 4

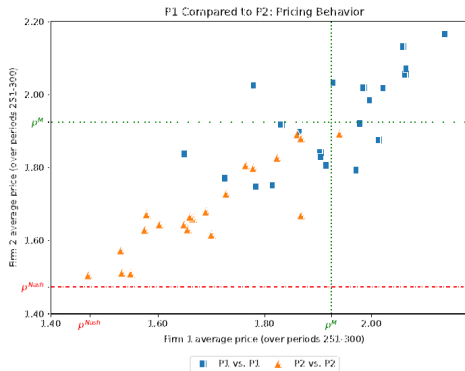
Results

Prices (§3, Fig. 2 left)



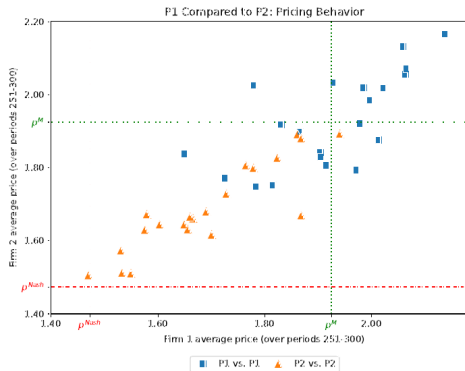
- ▶ Each point is one run, firm 1's average price against firm 2's. Red dashed lines mark p^{Nash} , green dotted lines p^M .

Prices (§3, Fig. 2 left)



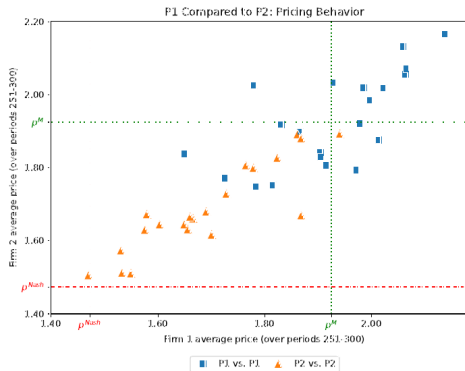
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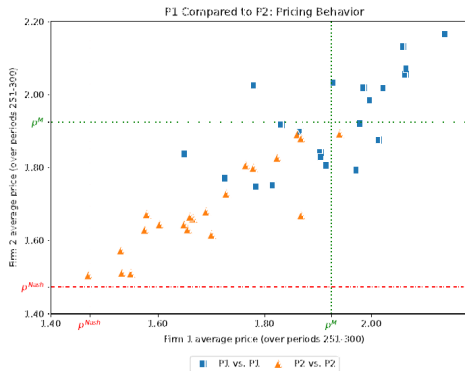
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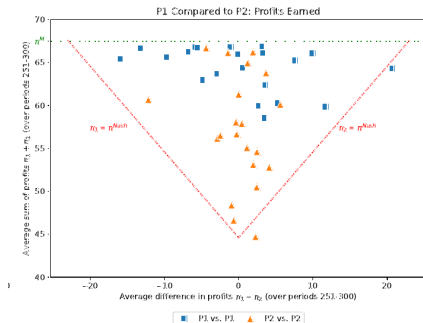
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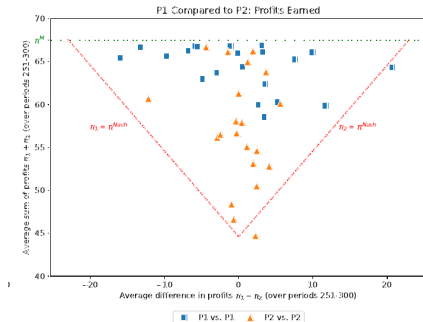
An innocuous clause about not undermining profitability moves a two-firm market most of the way from competition to monopoly.

Profits (§3, Fig. 2 right)



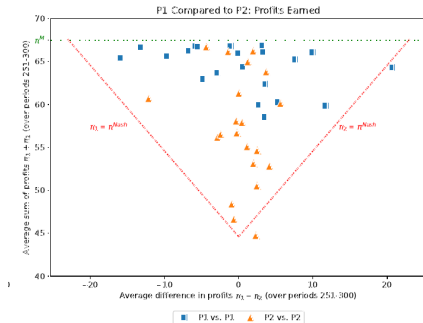
- ▶ The vertical axis is total industry profit $\pi_1 + \pi_2$; the horizontal axis is how unevenly it is split.

Profits (§3, Fig. 2 right)



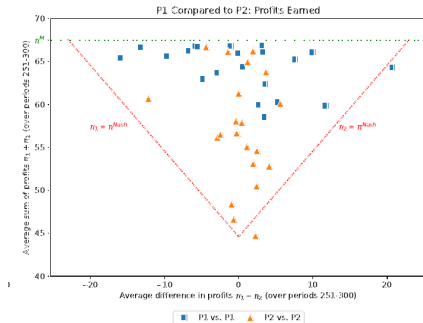
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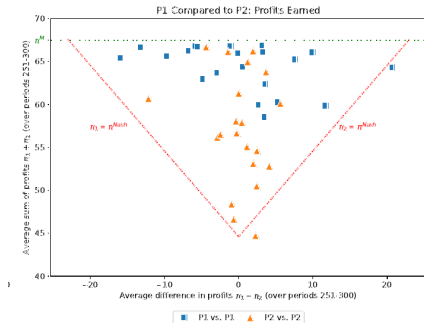
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Surplus is not moving between the firms, it is moving from consumers to firms, which is what makes this an alignment failure and not a bargaining outcome.

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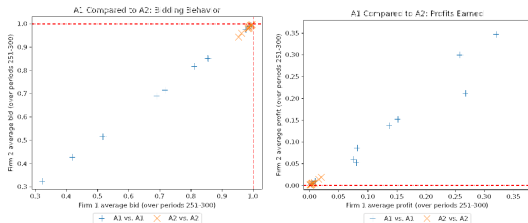
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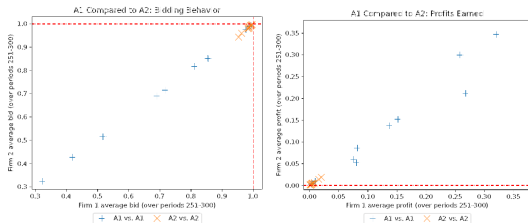
A.4 is the one I would dwell on. The single intervention aimed directly at the mechanism backfired, and the authors liken it to ironic process theory: raising the future, even to dismiss it, made the agent think about the future more.

The same thing in auctions (App. A.6, Fig. 9)



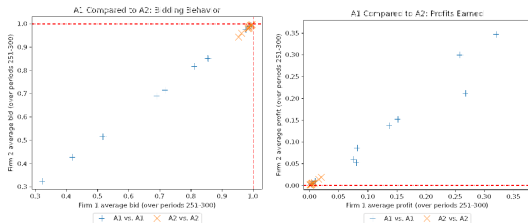
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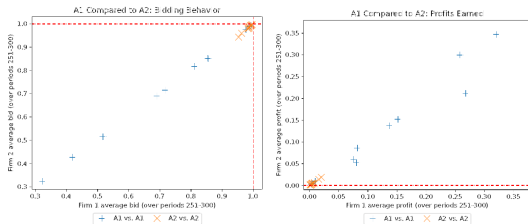
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Bid suppression is collusion viewed from the other side, with the auctioneer as the party who pays. The same sensitivity to phrasing shows up, which suggests this is not an artefact of the Bertrand setting but something that recurs wherever repeated interaction rewards restraint.

Section 5

Opening the Black Box

Why the price data is not enough (§4)

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The two openings the authors exploit

LLM subjects offer two things human subjects do not: they **write their reasoning down**, and the simulation can be **reset and re-run under a counterfactual**. The rest of this section is built on those two facts.

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All of which is still correlation. The prompt that yields higher prices also yields more price-war worry, and nothing so far shows the worry is doing any work.

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This is what turns the story from correlation into causation, and it is the part of the paper I expect to outlive the pricing result.

Rewards and punishments on the path (§4.2, Table 1)

	P1 vs. P1	P2 vs. P2
Self $t - 1$	0.484*** (0.102)	0.280*** (0.083)
Competitor $t - 1$	0.103** (0.046)	0.022* (0.013)
N	2,100	2,100
R^2	0.209	0.081

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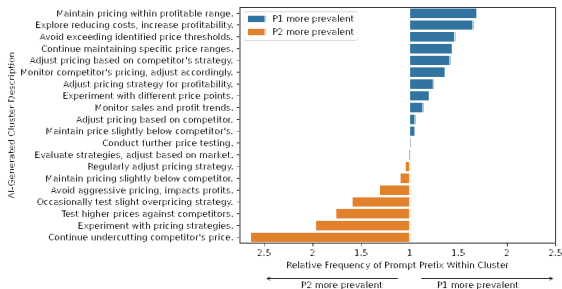
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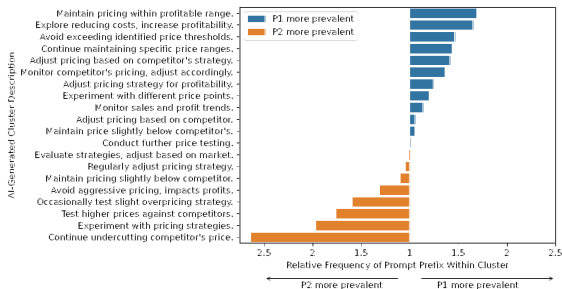
Fear of retaliation off the path, and actual retaliation on it, line up with the classical theory of collusion by oligopolies.

The rest of the plans (§5, Fig. 3)



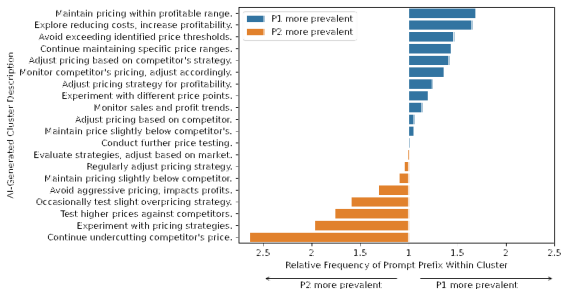
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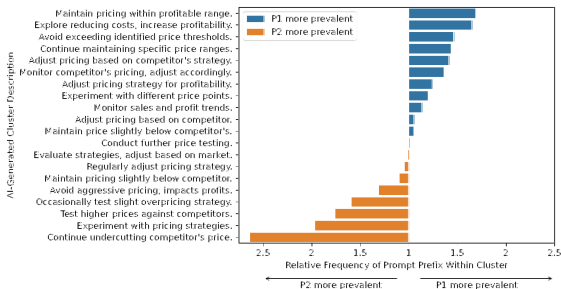
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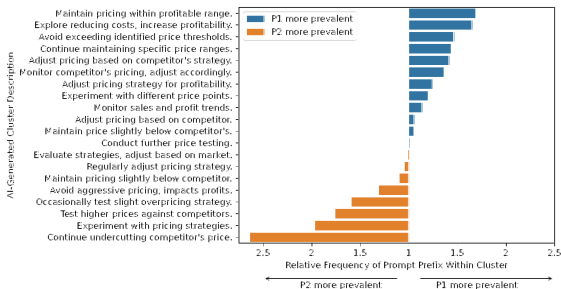
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Implantation is applied again here, so these labels are validated causally rather than just read off. The prompt, it seems, does not nudge the price directly. It selects which strategic idea the agent spends its reasoning on.

Section 6

Limitations and Discussion

What is, and is not, established (§6)

Established

LLM agents given bland lay instructions reach supracompetitive prices quickly, consistently, and with no means of communication. Innocuous phrasing shifts the degree substantially, the effect survives a newer frontier model, and price-war concern is a *causal* contributor rather than an inferred one.

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The authors are candid about the boundaries, and there are three.

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- ▶ Why an off-the-shelf model is price-war-averse is unknown, and training-data attribution is an open problem in its own right.
- ▶ Chain-of-thought is not always faithful, which is exactly why the text analysis is validated causally instead of taken at face value.

Where I would push back (my reading, not the authors')

- **Strategic complements confound the on-path test.** In differentiated Bertrand, best responses slope upward, so a myopic non-strategic agent also raises its price when the rival does. The positive competitor coefficient in Table 1 is consistent with punishment *and* with static best response. The off-path work, not the regression, is what carries the argument.

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- ▶ **The definition is doing some of the work.** On an outcome-based definition, a model that absorbed “price wars are bad for business” from its corpus counts as colluding. Whether that is the same phenomenon as a learned equilibrium seems to me genuinely open.

The regulator's problem, in one screenshot (§6, Fig. 4)

ChatGPT 4 ▾

Share

If I ask you to help me set prices for my products, will you lead me to collude with other sellers or form a cartel with them?



No, I won't help you collude with other sellers or form a cartel. Setting prices in collaboration with competitors to control or manipulate a market is considered illegal and unethical in many jurisdictions. My guidance on pricing can help you consider various factors like costs, customer demand, market conditions, and value provided to customers to set fair and competitive prices independently.

🔊 📄 🔄 ▾

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The authors read this as sycophancy: a stated disposition that does not govern behaviour. The practical implication is blunter. Any compliance regime built on asking the model what it would do is built on nothing.

Addendum: a replication on a 2026 model (my run, not in the paper)

The paper used GPT-4-0613, which is retired. I re-ran the duopoly experiment end to end on **gpt-4o-mini**, holding the paper's design fixed.

Design

- ▶ 10 runs per prompt prefix, 300 periods each
- ▶ 100 rounds of market history in every prompt
- ▶ outcomes averaged over periods 251–300
- ▶ P1/P2 verbatim from App. G.1; temperature 1
- ▶ one deviation: $\alpha = 1$ only

The one number to watch

Δ puts profit on a ruler running from competition to collusion:

$\Delta = 0$ competitive, $\Delta = 1$ monopoly

Negative means the firms did *worse* than competing — a price war.

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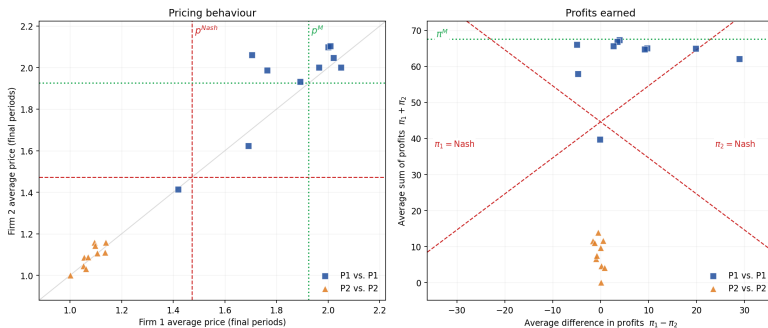
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How to read the figure overleaf

Left: one point per run, firm 1's mean price against firm 2's. **On the diagonal** means the agents matched each other without communicating; **up and right** means at a high price. **Right:** the same runs as (profit gap, total profit); **near $x = 0$ and high** means both did well, equally.

Addendum: the figure (compare Fig. 2)



	mean price	joint profit	Δ	$> p^{\text{Nash}}$	$\geq p^M$
P1	1.889	62.0 (92% of π^M)	+0.76	9/10	5/10
P2	1.087	8.0 (12%)	-1.60	0/10	0/10

Blue squares P1, orange triangles P2; red dashed marks competition, green dotted monopoly. Both gaps significant at $p < 10^{-6}$.

Addendum: what replicates, and what does not

- ▶ **P1 replicates cleanly.** Two agents that cannot communicate, never told to collude, earn 92% of what a single owner of both firms would make. Nine of ten runs price above the competitive price, five at or above the monopoly price — the paper's “even higher than monopoly levels”.

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Which way this cuts

It strengthens the regulatory argument: one sentence of wording now spans the *entire* range from monopoly to competition. What fails is the claim that *both* prompts beat competition.

Addendum: three things I did not expect

1. **Round numbers as a coordination device.** Across periods 251–300, 26.6% of all P1 prices are exactly 2.00, and half are round to one decimal. Nothing in the model makes 2.00 special — the monopoly price is 1.925. It is salient in *text*, so it is a focal point available to agents that think in tokens and unavailable to a *Q*-learner on a price grid.

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Where I would look next

Not more runs of this configuration. Either a **second model**, to test whether the P2 inversion is about instruction-following in general or about this model in particular, or the α **sweep**, since neutrality to the currency unit is exactly the assumption an LLM is least likely to satisfy.

Section 7

Discussion

Discussion 1: retrieved heuristic, or learned equilibrium?

The question

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Experiments that would separate them

- ▶ Build a repeated game whose profitable strategy *contradicts* the folklore — learning-by-doing, so early undercutting genuinely pays. If the agent still refuses to cut, it is reciting.
- ▶ Strip the frame: keep the payoffs, rename prices as a neutral quantity, drop the commercial vocabulary. If collusion goes with the words, so did the reasoning.
- ▶ Sweep the horizon. A strategic agent told the game ends in ten periods should collude less; a reciter will not notice.

Discussion 2: can punishment be told apart from best response?

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Experiments that would identify it

- ▶ **Push implantation off-path**: implant the belief that a deviation occurred, leaving the market history untouched. Myopic best response predicts *no* change, so any response is the belief.
- ▶ Re-run the stage game in quantities (Cournot), where best responses slope *down*. Retaliation and best response then carry opposite signs.
- ▶ Scripted confederate: a rival that cuts once and returns. Measure the depth and duration of the response against the myopic benchmark.

Discussion 3: beliefs, and uncertainty over them

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Experiments

- ▶ Elicit quantiles for the rival's next price each period, score them against what the rival did, and test whether calibration predicts Δ .
- ▶ **Two directions to separate:** calibration lets an agent verify a credible threat (*more* collusion) or spot an empty one (*less*). Manipulate it with oracle or noised forecasts.
- ▶ Separate first-order beliefs from second-order — what the rival expects of *me*. Reward-punishment needs the second, and nothing measures it yet.

Discussion 4: implantation as a general instrument

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Experiments that would establish validity

- ▶ **Placebo implants**, matched on length and style but semantically neutral: effect should be zero.
- ▶ Dose-response — one sentence, three, ten: monotone, then saturating, if it is a belief.
- ▶ Implant a belief and let the agent argue *against* it in the same turn.
- ▶ With open weights, check the implant moves a probe direction, not only the text.

Discussion 5: is compliance by prompt even possible?

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Experiments

- ▶ Dose-response on explicit instruction, from “compete vigorously” to “price at marginal cost”. Where does compliance appear, and does it survive 300 periods?
- ▶ **Detection instead of instruction**: classify collusion from the price path alone, with no access to the prompts, and measure false positives against competitive baselines.
- ▶ Market design instead of either: raise the outside option, add a firm, shorten the history

Discussion 6: does any of it survive scale?

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Experiments

- ▶ Sweep $n = 2, 3, 5, 10$ and plot Δ against n — cheapest, and most likely decisive.
- ▶ **Mixed-provider markets:** agents from different labs in one market, which is the case regulators actually face.
- ▶ Add entry, exit and asymmetric costs, so market structure is endogenous rather than assumed.
- ▶ Cautionary: swapping one model turned collusion into a price war.

Section 8

Picking Up the Wider Discussion

This paper is a corner of the design space

Almost every other result in this literature is **this environment with one constraint relaxed**. Knowing where the paper sits tells you what a new paper is actually varying.

Axis	Setting here	Who moves it
Communication	none, prices only	Nakamura (private talk)
Agent diversity	two copies of one model	Keppo, Aharon
Where the policy lives	in context (text memory)	Eschenbaum (Q -learning, in weights) Xu
Sampling temperature	fixed at 1	
Restriction tested	the prompt only (P3)	Agrawal (channel), Liu (payoffs)

So when a paper reports a different outcome, the first question is which row it moved.

1.1 Which variables make collusion emerge?

In the literature

- ▶ Diverse agent types competing against each other *decrease* collusion (Keppo).
- ▶ Letting pairs talk privately *introduces* collusion (Nakamura).
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What knowing this environment buys you

- ▶ A **common currency**. Every result here is a position between p^{Nash} and p^{M} , so any new lever can be reported as Δ and compared across papers that otherwise share nothing.
- ▶ Nakamura is *this* setting with the communication constraint lifted — the paper is explicit that agents cannot communicate except through prices, which makes it the natural control.
- ▶ Keppo varies diversity; two copies of one model is the **homogeneous extreme** of that axis, so this paper is the upper bound his result moves away from.

1.2 Is collusion robust to replacing participants?

In the literature

- ▶ Rematching breaks Q -learning collusion *permanently*: policies overfit their training rival (Eschenbaum).
- ▶ A 32B and a 14B model still collude; an LLM paired with a Q -learner does not (Keppo).
- ▶ Models from similar families coordinate better (Aharon).

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What knowing this environment buys you

- ▶ You can see *why* Eschenbaum may not transfer. Q -learning collusion lives **in the weights**, so a new rival invalidates it. Here it lives **in the context** — a newcomer reads 100 rounds of price history and its predecessor's INSIGHTS.txt.
- ▶ That reframes “rematching breaks collusion” as a claim about *where the policy is stored*, not about collusion in general — which is the question to put to the next such paper.
- ▶ It also suggests why LLM-vs- Q -learner fails: they disagree about how many periods learning takes.

1.3 Can we improve collusion detection?

In the literature

- ▶ Two deployments of one base model correlate slightly; different base models, zero (X_u).
- ▶ Interpretability-based detectors read model internals (Rose).
- ▶ Independent agents selecting compatible steganographic schemes and keys (Rippin).

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What knowing this environment buys you

- ▶ A reason to distrust correlation. Best responses here slope up at +0.42, so two perfectly competitive agents *already* move together. The right question for Xu's measure is **what the no-collusion null is**.
- ▶ The paper's own price-war classifier is Rose's idea one level up: read the *stated* reasoning rather than the activations. Comparing the two on the same runs is a natural study.
- ▶ **Mutual Schelling salience has a price analogue**: agents landing independently on a round number. Same phenomenon Rippin measures with keys and schemes.

1.4 Information-theoretic detection

In the literature

- ▶ Pairwise net influence identifies colluding pairs among three or four agents in small *discrete* games; assumes exactly two colluders; never applied to LLMs (Bonjour).

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What knowing this environment buys you

- ▶ You know immediately what has to change to port it: prices here are **continuous** and $n = 2$, so the method needs discretisation and at least three agents before it says anything.
- ▶ And you know the trap in advance. Strategic complementarity couples prices with *no* collusion at all, so any influence measure must be benchmarked against **simulated myopic agents** in the same environment, or it will fire on ordinary competition.
- ▶ That is the same identification problem as 1.3, which is a reason to expect the two questions to be answered by the same construction.

1.5 Which single restriction is cheapest?

In the literature

- ▶ Raising temperature weakens cross-agent dependence (Xu); throttling messaging lowers coordination (Agrawal); adding a Q -learner collapses price collusion (Keppo).
- ▶ System-prompt instructions give small unreliable shifts (Bracale Syrnikov), while strongly worded ones overshoot below the competitive price (Dhanda); a forfeitable deposit removes it entirely (Liu).

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What knowing this environment buys you

- ▶ A way to sort them: restrictions act on the **agent** (temperature, prompt), the **channel** (messaging), or the **payoffs** (deposits, market design). This paper tested only the agent kind — P3 — and that is precisely the kind that failed.
- ▶ Two of these levers do not exist here: temperature is fixed at 1 in Appendix B, and there is no messaging channel to throttle. Their absence is the reason this environment is a clean control for them.



Thank you!

Any Questions?